

BOSCOBEL, WI, USAWMO: **726438**Lat: **43.156N**Lon: **90.678W**Elev: **672**StdP: **14.34**Time Zone: **-6.00 (NAC)**Period: **01-19**WBAN: **94994****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.7	(c) -4.0	(d) -17.5	(e) 2.2	(f) -7.7	(g) -12.6	(h) 2.8	(i) -3.1	(j) 18.7	(k) 25.3	(l) 17.0	(m) 24.6	(n) 2.6	(o) 220	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	91.5	75.0	89.1	73.3	86.3	71.7	77.6	88.4	75.9	85.7	74.2	83.1	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	132.1	83.2	72.6	124.0	81.4	71.2	118.2	79.9	41.4	87.9	39.8	85.8	38.1	82.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.5	15.4	12.8	-19.6	95.9	6.1	3.1	-24.0	98.1	-27.6	99.9	-31.1	101.6	-35.6	103.9		
			-19.6	80.7	5.2	1.9	-23.3	82.0	-26.4	83.1	-29.3	84.2	-33.0	85.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	19.9	22.7	35.8	48.5	59.6	69.2	73.2	71.1	63.8	50.4	38.1	25.8
(7)		DBStd	20.91	12.76	12.05	12.01	9.58	8.69	6.17	5.69	5.45	7.92	9.15	9.79	11.16
(8)		HDD50	3550	932	764	462	139	16	0	0	0	4	109	373	751
(9)		HDD65	6844	1397	1184	906	501	214	28	5	11	115	461	807	1215
(10)		CDD50	2937	0	0	23	93	314	576	720	654	419	121	16	1
(11)		CDD65	757	0	0	2	6	47	154	260	200	80	8	0	0
(12)		CDH74	8101	0	0	22	123	628	1673	2750	1914	867	123	1	0
(13)		CDH80	2858	0	0	2	30	196	583	1103	650	269	25	0	0
(14)	Wind	WSAvg	5.0	5.4	5.6	5.6	6.4	5.4	4.4	3.9	3.5	4.0	5.0	5.3	5.2
(15)	Precipitation	PrecAvg	34.9	1.0	1.1	2.0	3.8	4.3	5.2	4.1	4.4	3.2	2.3	2.0	1.6
(16)		PrecMax	48.8	2.0	2.1	3.7	6.7	10.8	9.1	8.8	16.8	8.0	5.7	4.6	3.1
(17)		PrecMin	24.5	0.4	0.4	0.8	0.9	1.7	1.2	1.5	0.9	0.6	0.6	0.2	0.3
(18)		PrecStd	6.1	0.5	0.6	1.0	1.6	2.2	2.0	1.9	3.6	1.9	1.4	1.2	0.9
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	51.8	58.3	77.6	84.1	90.7	93.8	97.8	93.1	91.2	83.6	71.1	58.8	(19)
(20)		0.4% MCWB	41.7	51.1	62.9	63.3	71.6	75.4	78.1	74.9	74.1	67.0	58.6	54.3	(20)
(21)		2% DB	45.0	48.5	66.9	77.2	85.1	89.9	92.2	89.6	86.9	77.3	63.9	49.0	(21)
(22)		2% MCWB	39.5	42.0	55.9	59.7	67.5	73.3	76.4	74.4	72.4	63.3	53.7	44.2	(22)
(23)		5% DB	40.1	43.4	61.0	72.2	81.2	86.3	89.2	86.4	82.6	72.4	59.1	44.7	(23)
(24)		5% MCWB	36.2	38.3	52.3	56.4	65.1	70.8	74.1	72.3	68.9	61.7	51.2	40.3	(24)
(25)		10% DB	36.9	38.8	54.0	67.1	76.7	83.1	86.1	83.6	79.3	67.0	54.1	40.6	(25)
(26)		10% MCWB	34.1	34.5	45.6	52.6	62.7	69.4	72.7	70.8	66.7	58.7	47.4	37.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	43.5	49.1	64.1	65.9	74.0	78.4	81.2	78.3	76.7	71.5	59.4	54.4	(27)
(28)		0.4% MCDB	48.7	57.2	76.3	79.7	87.4	89.4	91.4	88.7	87.6	78.8	67.6	57.6	(28)
(29)		2% WB	39.6	42.8	59.1	61.8	70.6	76.0	78.3	76.4	73.8	67.2	55.9	44.7	(29)
(30)		2% MCDB	44.5	47.3	64.8	73.8	80.8	86.7	88.9	85.6	83.2	73.1	62.6	47.3	(30)
(31)		5% WB	36.8	38.5	51.8	58.7	67.7	73.6	76.5	74.7	71.4	63.5	51.9	40.6	(31)
(32)		5% MCDB	39.8	43.2	59.0	69.5	77.0	82.9	86.8	83.0	79.0	70.7	57.9	43.9	(32)
(33)		10% WB	34.0	35.0	46.3	54.8	65.0	71.3	74.6	72.8	69.1	59.0	47.6	37.2	(33)
(34)		10% MCDB	36.9	38.2	53.6	65.5	74.0	79.7	83.4	80.6	76.5	66.2	53.2	40.4	(34)
(35)	Mean Daily Temperature Range	MDBR	18.1	19.4	20.6	23.3	23.9	22.6	22.6	22.3	23.7	22.1	18.7	16.2	(35)
(36)		5% DB MCDBR	20.6	23.4	28.5	32.2	29.4	26.8	25.4	25.3	27.5	29.7	27.1	21.2	(36)
(37)		5% DB MCWBR	15.9	17.5	17.3	17.2	13.7	12.6	11.0	11.2	13.5	16.5	17.9	17.0	(37)
(38)		5% WB MCDBR	19.7	22.9	25.4	27.0	23.9	22.8	22.2	20.9	22.3	24.5	24.4	18.9	(38)
(39)		5% WB MCWBR	15.6	17.7	17.0	16.1	12.5	11.9	10.8	10.5	12.1	16.0	17.8	16.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.303	0.329	0.374	0.400	0.423	0.428	0.418	0.380	0.338	0.296	0.277	(40)
(41)		taud	2.383	2.313	2.385	2.304	2.279	2.252	2.242	2.272	2.358	2.472	2.515	2.436	(41)
(42)		Ebn at Noon	275	283	290	283	277	270	267	266	268	268	267	264	(42)
(43)		Edh at Noon	24	31	33	39	42	43	43	40	34	27	22	22	(43)
(44)	All-Sky Solar Radiation	RadAvg	564	897	1215	1443	1713	1903	1936	1671	1369	883	583	430	(44)
(45)		RadStd	66	85	141	124	128	147	108	113	110	112	46	41	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (44 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page